

BCD Process & Cadence Tutorial

ET4382

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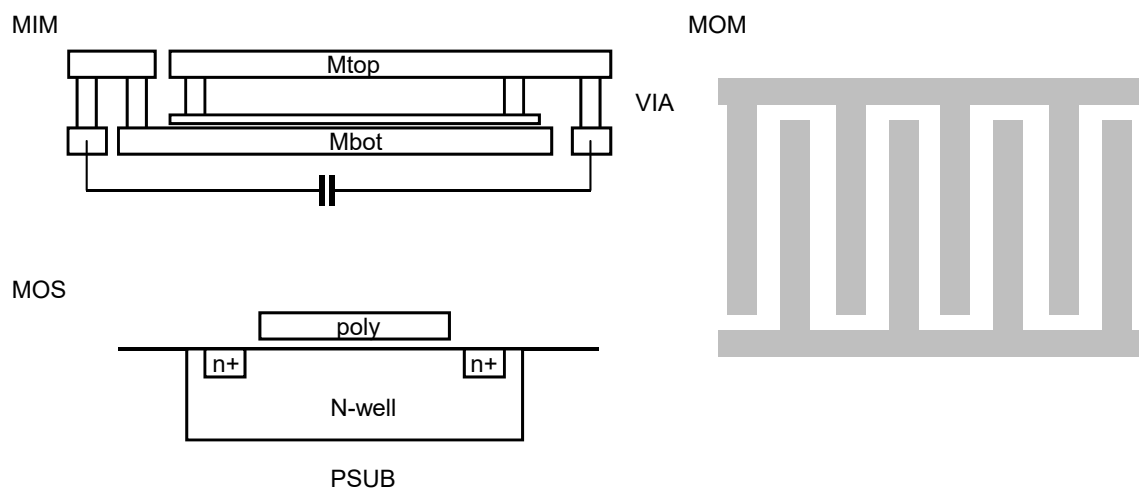
Agenda

- Introduction to BCD process
 - References
 - Process Design Rule Manual
 - Alan Hastings, *The art of analog layout*, 2006.
- Cadence Q&A
 - Need to sign Statement of Confidentiality first

BCD Process

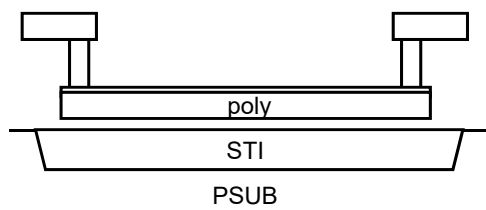
- Bipolar
 - NPN, PNP
- CMOS
 - HV Isolation
- DMOS
- Diodes
 - PN junction
 - Zener diode
 - Schottky diode
- Capacitors
 - MIM
 - MOM
 - MOS
- Resistors
 - Poly
 - N-well
 - Metal

Capacitors

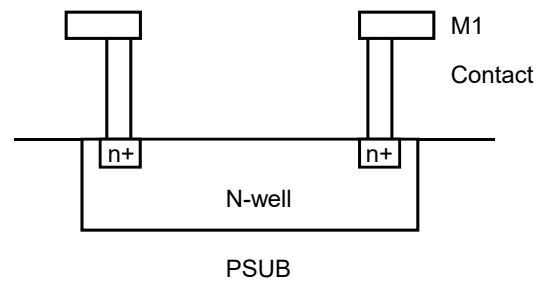


Resistors

- Poly
 - Silicided/unsilicided
 - Temperature coefficient

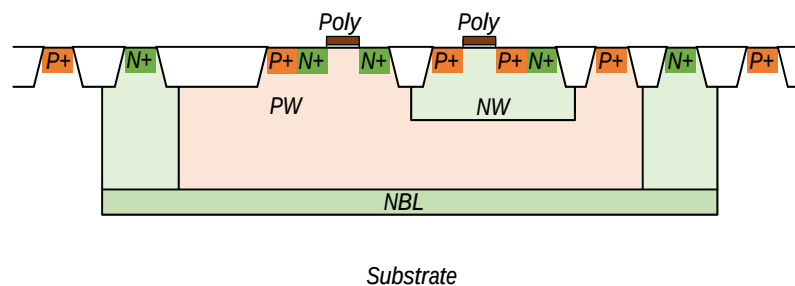


- N-well
 - Nonlinear



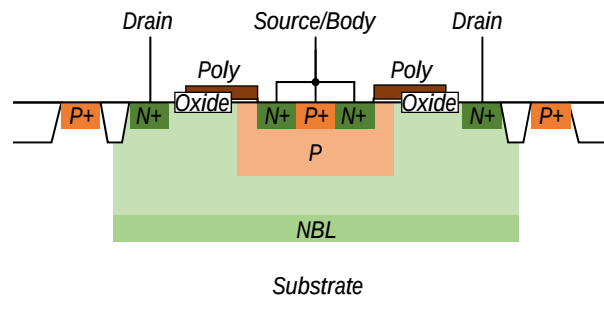
CMOS

- High voltage n-well solution

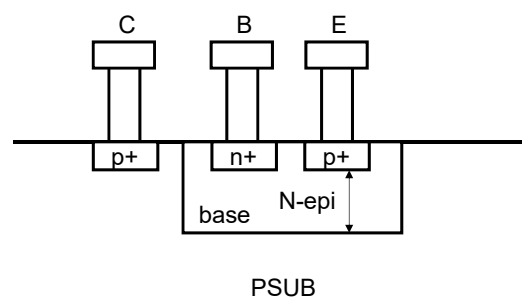


DMOS

- High voltage MOS transistors
- $V_{DS,max} > V_{GS,max}$

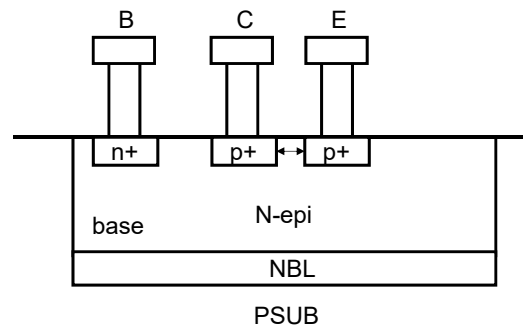


Substrate PNP

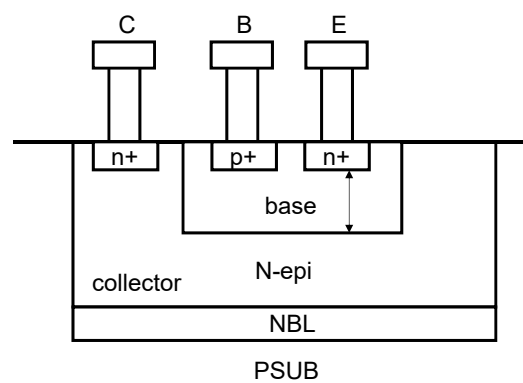


Collector always tied to substrate!

Lateral PNP

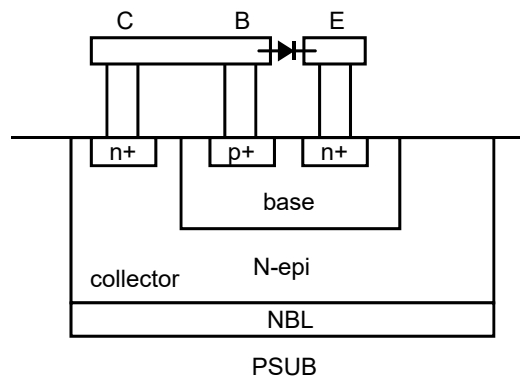


Vertical NPN



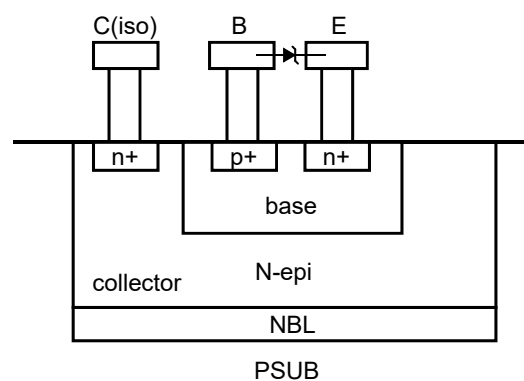
Diodes

- Why not use a simple PN junction?
- PN junction (diode-connected NPN)



Zener Diode

- Zener breakdown
- Use in reverse bias
- Base-emitter junction of NPN
- Tie iso to E or higher voltage
 - Why not short C and B?



Schottky Diode

- ~0.4V forward voltage

